

DC Motor Bidirectional Control With TV Remote

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This project describes how a DC motor can be moved in forward or reverse direction using a TV

or DVD remote control. The goal is to construct a simple bi-directional motor driver that uses modulated

infrared (IR) 38kHz pulse train for the purpose without using any microcontroller or programming.

The author's prototype is shown in Fig. 1.

Circuit and working

Circuit diagram of the project is shown in Fig. 2. It is built around IR receiver module TSOP1738 (IRRX1), decade counter 4017B (IC2), motor driver L293D (IC3), PNP transistor BC557 (T1), two BC547 NPN transistors (T2 and T3), 5V regulated power supply (IC1), and a 9V battery.

The 9V battery is connected through diode D1 to voltage regulator 7805 to generate 5V DC required for



Fig. 1: Author's prototype

Truth Table of L293D Motor Driver Table 1

Cases	IN1	IN2	Motor status
O0 - high	1	1	Off state
O1 - high	0	1	Forward direction
O2 - high	1	1	Off state
O3 - high	1	0	Reverse direction
O4 - high (Reset)	1	1	Off state

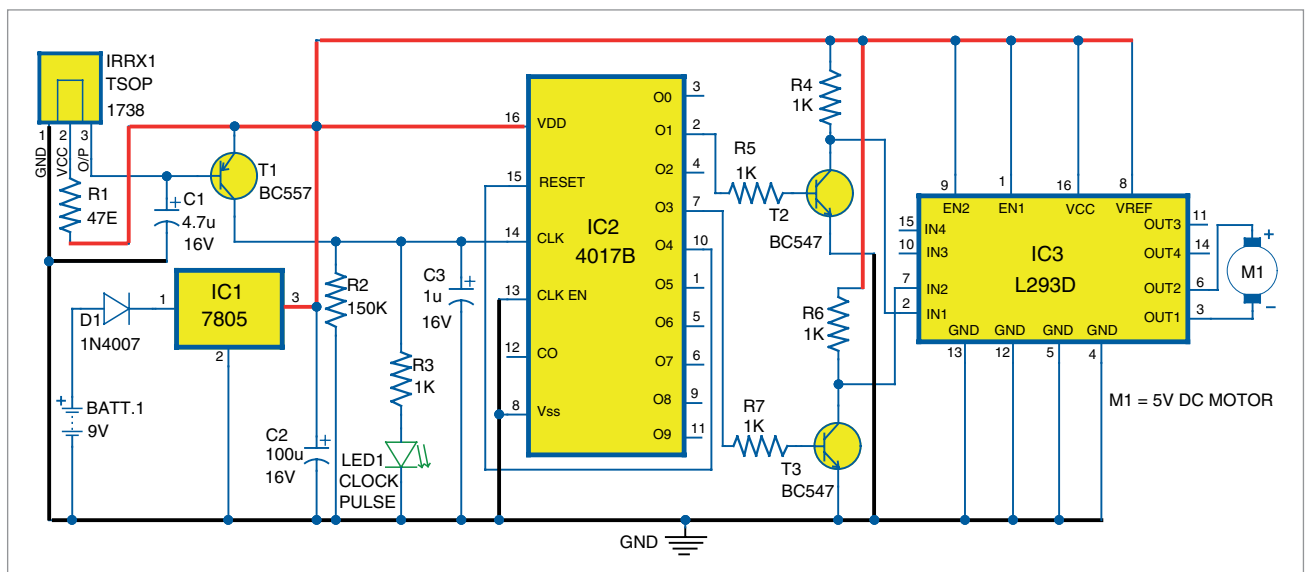


Fig. 2: Circuit diagram of the DC motor driver